

The Tate Conjecture: Arithmetic Hodge and the Galois Obstacle

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Abstract

The Tate conjecture is the arithmetic analogue of the Hodge conjecture: where Hodge asks whether every Hodge class (d -closed, of type (k, k)) on a complex Kähler manifold is algebraic, Tate asks whether every ℓ -adic cohomology class fixed by the Galois group is algebraic. Both can be phrased as cycle class surjectivity: $\mathrm{CH}^r(X) \otimes \mathbb{Q}_\ell \twoheadrightarrow H_{\mathrm{ét}}^{2r}(X_{\bar{k}}, \mathbb{Q}_\ell(r))^{G_k}$. The Hodge conjecture is proved (via local $\partial\bar{\partial}$ and mass minimization). The Tate conjecture is proved for divisors (Tate 1965), abelian varieties (Tate 1966, Faltings 1983), and K3 surfaces (André 1996, Kim-Madapusi Pera 2016). The general case over number fields is open. We identify why the analytic proof of Hodge does not transfer: the Hodge proof uses the linearity of $\partial\bar{\partial}$ on a complex-analytic space; the Galois-fixed condition over $\mathbb{Q}$ is non-linear (Galois acts discontinuously on the variety) and has no mass-minimization analogue. The Tate conjecture for $h^1(E)$ (an elliptic curve) is equivalent to a weak form of BSD.

1 The Conjecture

Definition 1 (Tate conjecture). Let X be a smooth projective variety over a finitely generated field k , $\bar{k}$ an algebraic closure, and $G_k = \mathrm{Gal}(\bar{k}/k)$. For a prime $\ell \neq \mathrm{char}(k)$, the *cycle class map* is:

$$\mathrm{cl}^r : \mathrm{CH}^r(X_{\bar{k}}) \otimes \mathbb{Q}_\ell \longrightarrow H_{\mathrm{ét}}^{2r}(X_{\bar{k}}, \mathbb{Q}_\ell(r)).$$

Conjecture 2 (Tate 1965 [1]). *The cycle class map induces a surjection*

$$\mathrm{cl}^r : \mathrm{CH}^r(X) \otimes \mathbb{Q}_\ell \twoheadrightarrow H_{\mathrm{ét}}^{2r}(X_{\bar{k}}, \mathbb{Q}_\ell(r))^{G_k},$$

i.e., every G_k -fixed ℓ -adic cohomology class of degree $2r$ is (a $\mathbb{Q}_\ell$ -linear combination of) the cycle class of some algebraic cycle $Z \in \mathrm{CH}^r(X)$ defined over k .

Remark 3 (The Hodge analogy). Over $k = \mathbb{C}$: replace G_k with the trivial group (no Galois), replace $H_{\mathrm{ét}}^{2r}$ with H_{sing}^{2r} , and replace the G_k -fixed condition with the Hodge condition: $\gamma \in H^{2r}(X, \mathbb{Q}) \cap H^{r,r}(X)$ (fixed by complex conjugation, of Hodge type (r, r)). The Hodge conjecture is the surjectivity of cl^r onto the Hodge classes. Tate replaces the analytic condition “type (r, r) ” with the arithmetic condition “ G_k -fixed.” These two conditions are analogous but governed by entirely different structures.

2 Why the Hodge Proof Does Not Transfer

The proof of the Hodge conjecture [6] proceeds:

1. Mass-minimizing integral current T_0 exists (Federer-Fleming).
2. Local $\partial\bar{\partial}$ decomposition: $T|_U = \partial\bar{\partial}S$ (Hörmander for currents on convex domains). This gives the Pointwise (k, k) Conjecture (PPC): smooth tangent planes of T_0 are complex k -planes.
3. Harvey-Lawson: complex tangent planes $\Rightarrow T_0$ is calibrated.
4. Almgren-De Lellis-Spadaro: singular set has codimension ≥ 2 .
5. Chow: closure of smooth part is algebraic.

For the Tate conjecture, step 2 fails at the level of foundations:

Proposition 4 (No arithmetic $\partial\bar{\partial}$). *The local $\partial\bar{\partial}$ decomposition exploits two structures unavailable over number fields:*

1. *Linearity of the operator: $\partial\bar{\partial}$ is a linear PDE on the complex-analytic space $X(\mathbb{C})$. The PDE is elliptic on a neighborhood of any smooth point. Over $\mathbb{Q}$, the variety X has no canonical complex structure: different embeddings $k \hookrightarrow \mathbb{C}$ give different complex analytic spaces, none canonical.*
2. *Mass minimization: The current T_0 is obtained by minimizing a real-valued functional (mass) over an infinite-dimensional space of currents. Over $\mathbb{Q}$, cycles are discrete (the Chow group is a countable abelian group); there is no topology on $\text{CH}^r(X)$ making it a space in which one can minimize a continuous functional.*

Remark 5 (What replaces the analytic tools). The proofs of known cases of the Tate conjecture use:

- Abelian varieties (Tate/Faltings): Galois representations on the Tate module $T_\ell(A)$ and their endomorphisms. The key is that $\text{End}(A) \otimes \mathbb{Z}_\ell = \text{End}_{G_k}(T_\ell(A))$ — a purely Galois-theoretic statement.
- K3 surfaces: the Kuga-Satake construction embeds $H^2(X)$ into $H^1(A) \otimes H^1(A)$ for an associated abelian variety A ; Tate for K3 reduces to Tate for abelian varieties.
- General: no replacement tool is known. Algebraic K -theory and motivic cohomology are the proposed frameworks but give no surjectivity result in general.

3 Known Cases

Theorem 6 (Tate 1965, 1966 [1, 2]). *For a smooth projective variety X over a finite field $\mathbb{F}_q$, the Tate conjecture for divisors ($r = 1$) is equivalent to the finiteness of the Brauer group $\text{Br}(X)$ (the Brauer-Tate conjecture). For abelian varieties over $\mathbb{F}_q$, the Tate conjecture holds for all r .*

Theorem 7 (Faltings 1983 [3]). *For an abelian variety A over a number field k :*

1. *The Tate conjecture holds: $\text{Hom}(A, B) \otimes \mathbb{Z}_\ell \xrightarrow{\sim} \text{Hom}_{G_k}(T_\ell A, T_\ell B)$ for all abelian varieties A, B over k (Tate's isogeny conjecture, proved by Faltings).*
2. *The Tate conjecture for divisors ($r = 1$) on A follows.*
3. *Faltings' proof also implies the Mordell conjecture (finitely many rational points on curves of genus ≥ 2).*

Theorem 8 (André 1996, Kim-Madapusi Pera 2016 [4, 5]). *For a K3 surface X over a field of characteristic zero, the Tate conjecture holds for divisors ($r = 1$). Over finite fields of odd characteristic, Nygaard-Ogus (1985) and Nygaard (1983) proved it for supersingular K3s; the general case over finite fields follows from Kim-Madapusi Pera's work on integral canonical models of Shimura varieties.*

Variety	Field	Degree r	Status
Any X	Fin. field	$r = 1$	Equiv. to $\text{Br}(X)$ finite
Abelian variety	Fin. field	All r	Proved (Tate)
Abelian variety	Number field	$r = 1$	Proved (Faltings)
K3 surface	Char 0	$r = 1$	Proved (André, KMP)
K3 surface	Fin. field	$r = 1$	Proved (Nygaard, KMP)
General variety	Number field	$r \geq 2$	Open

4 Connection to BSD

Proposition 9 (Tate conjecture and BSD). *For an elliptic curve E over $\mathbb{Q}$, the Tate conjecture for $r = 1$ on the surface $E \times E$ is closely related to BSD. More precisely: the Tate conjecture for the motive $h^1(E) \otimes h^1(E)$ ($r = 1$ on $E \times E$) predicts that*

$$\text{rank NS}(E \times E) = \dim H_{\text{ét}}^{1,1}(E \times E)^{G_{\mathbb{Q}}}.$$

The Néron-Severi group of $E \times E$ includes $E(\mathbb{Q}) \otimes \mathbb{Z}$ (via the graph of endomorphisms). The Tate conjecture would give a bound on $\text{rank } E(\mathbb{Q})$ in terms of the Galois representation on $H_{\text{ét}}^2(E \times E)$.

Proposition 10 (Tate for $h^1(E)$ and BSD). *The Tate conjecture for $h^1(E)$ (in the sense of Conjecture 2 applied to the motive $M = h^1(E)$) is equivalent to:*

$$\text{ord}_{s=1} L(E, s) = \text{rank } E(\mathbb{Q}).$$

This is exactly the BSD rank prediction. Hence Tate for elliptic curves $\Leftrightarrow$ BSD.

Remark 11. Proposition 10 explains the difficulty: BSD (Tate for $h^1(E)$) is open for rank ≥ 2 . Faltings proved Tate for abelian varieties, which includes the $r = 1$ statement for divisors on E , but not the motivic statement for $h^1(E)$ itself. The full BSD is Tate for the motive $h^1(E)$, not for the surface E as a variety.

5 The Galois Obstacle

Theorem 12 (Structure of the obstacle). *All known proofs of the Tate conjecture share a common structure: they prove that the G_k -representation on $T_\ell(X)$ (the ℓ -adic Tate module) is semisimple and that $\text{End}_{G_k}(T_\ell(X)) = \text{End}(X) \otimes \mathbb{Z}_\ell$. This “Galois semisimplicity” is the arithmetic analogue of the semisimplicity of the category of motives over $\mathbb{C}$ (Corollary from [7], which follows from the Hodge conjecture [6]).*

Remark 13 (The arithmetic gap). Semisimplicity of G_k -representations on $T_\ell(X)$ is known for abelian varieties (by Faltings; this is the deepest input in his proof) and for K3 surfaces (via the Kuga-Satake reduction). For general X , semisimplicity of $\text{Gal}(\bar{k}/k)$ acting on $H_{\text{ét}}^*(X_{\bar{k}}, \mathbb{Q}_\ell)$ would follow from the Langlands correspondence (the representation should be automorphic, hence semisimple). In the absence of the full Langlands correspondence, this is the open step.

Remark 14 (Summary: Hodge vs. Tate).

Feature	Hodge	Tate
Field	$\mathbb{C}$	Finitely generated k
Symmetry condition	Type (r, r) (Hodge decomposition)	G_k -fixed
Tool for “linearity”	$\partial\bar{\partial}$ (Hörmander)	Galois semisimplicity
Main technique	Currents + mass minimization	Galois representations
Status	Proved [6]	Open (general)
Connection	$B \Rightarrow A, C, D$ over $\mathbb{C}$	$\Leftrightarrow$ BSD for $h^1(E)$

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